

DESIGN FOR ENVIRONMENT

A product impacts the environment across its lifecycle—starting from manufacturing to disposal. Designers can attempt to reduce the overall impact at the design stage itself. Here is how



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With increasing awareness about environment protection, every new product is expected to meet plethora of environment-friendly standards. While multitude of standards are available, there is little information available in the public domain for the designers on how to meet the environmental standards when designing a product. Engineers face twin challenges of meeting the environmental standards for the new products that they

design as well as making some of the existing products meet the standards. Both cases pose their own challenges that the designers have to meet.

This article serves as a small introduction for the design engineers in meeting the challenges. It elucidates the design aspects to meet the environmental standards. Before getting into the details, however, the designers should understand that the products today are not addressed in a standalone fashion, i.e., the designers have to worry not only about the final user of the product but all

the users of the product in its life, typically this is called as product life-cycle (PLC). Also, there are very few vendors who manufacture products in-house. Most of them outsource the manufacturing to contract manufacturers who are known as electronics manufacturing service (EMS) vendors.

Designers should be aware that when it comes to environment, each country has its own standard. While the basic philosophy of design remains more or less the same, the limits of compliance to standards vary from country to country. In the specific context of India today, the only environmental standard that is being insisted for electronic products is the star rating of Bureau of Energy Efficiency (BEE) wherein all electrical and electronic products are rated on a 5-star scale; the more the numbers of stars, the higher the energy efficiency of the product.

Anatomy of product lifecycle

All products typically go through a four-phase cycle before these are withdrawn.

Fig. 1 shows the conventional lifecycle where after the use (end of the product's active life), the product invariably gets discarded. So far this had been the practice till the discarded materials called e-waste started polluting the environment. There is now a greater emphasis on sustainable eco-friendly products, which typically means the products have to be environment-friendly and easily recyclable when their life is over. The product lifecycle to meet this environmental requirement is shown in Fig. 2. The key